Quiz (9) MCS 320 Jan Verschelde Friday 31 July 2026

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

For some parameter s, consider the space curve $(x(s,t), y(s,t), z(s,t))$ with its coordinates defined by

$$x(s,t) = r(s,t)\cos(4t), \quad y(s,t) = r(s,t)\sin(4t), \quad z(s,t) = r(s,t)\sin(st),$$

where $r(s,t) = 2 + 4/5\cos(st)$, for t ranging from 0 to 2π.

Write code to define an animation with eleven frames.

The frames are defined by the value for s, which ranges from 1 to 2.